



SYNTHESIS PROJECT

An AprilTag-Based Approach to Trajectory Recording and Relocalization for Mobile Robots

Joao Pedro Martins do Lago Reis and Yann Kelvem da Silva Ramos

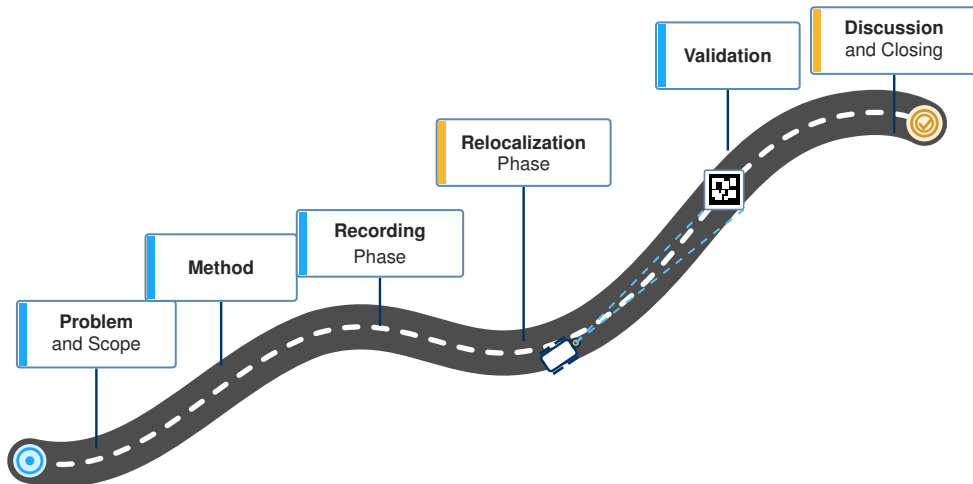
Master's degree in artificial perception and robotics

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Supervisor: Dr. Sébastien Lengagne

Evaluator: Dr. Aufrère Romuald

Outline



Problem and Scope

For whom

Polytech practical-work students using
AgileX LIMO in indoor AprilTag setups.

Problem and Scope

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Why

Need a reproducible workflow to record a path and recover pose relative to it for control-law exercises.

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Scope

In: offline reconstruction and relocalization outputs (D , ϕ).

Out: control-law and trajectory replay.

Method

Three-Stage Method



1) Record
detections dataset

Main Equation

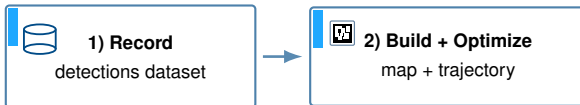
Recording

$${}^{\text{cam}}T_{\text{tag}}(t)$$

Artifacts

Method

Three-Stage Method



Main Equation

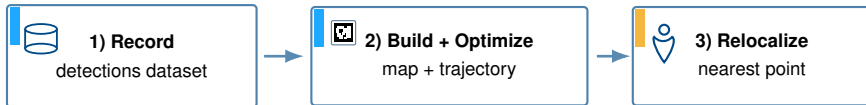
Build + Optimize

$$\min_{\{w_{tag_i}\}} \sum_{(i,j) \in \mathcal{E}} \rho(\|err_{ij}\|^2)$$

Artifacts

Method

Three-Stage Method



Main Equation

Relocalization

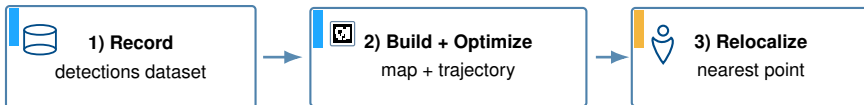
$$D = \|\mathbf{p}_{\text{cur}} - \mathbf{p}_{\text{nearest}}\|$$

$$\phi = \text{wrap}(\psi_{\text{cur}} - \psi_{\text{nearest}})$$

Artifacts

Method

Three-Stage Method



Main Equation

Relocalization

$$D = \|\mathbf{p}_{\text{cur}} - \mathbf{p}_{\text{nearest}}\|$$

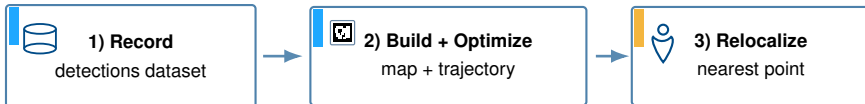
$$\phi = \text{wrap}(\psi_{\text{cur}} - \psi_{\text{nearest}})$$

Artifacts

 tag_map.yaml

Method

Three-Stage Method



Main Equation

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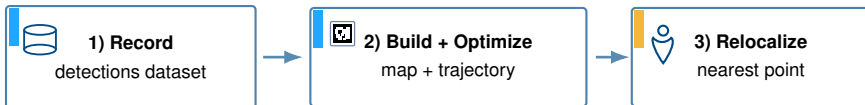
tag_map.yaml



trajetoria_camera.csv

Method

Three-Stage Method



Main Equation

Relocalization

$$D = \|\mathbf{p}_{\text{cur}} - \mathbf{p}_{\text{nearest}}\|$$

$$\phi = \text{wrap}(\psi_{\text{cur}} - \psi_{\text{nearest}})$$

Artifacts

 tag_map.yaml

 trajetoria_camera.csv

 D [m], ϕ [deg]

Recording Phase

Recording Phase — Template

To fill later:

- ▶ Inputs and sensors
- ▶ Offline optimization steps
- ▶ Generated files
- ▶ Key interpretation points

Relocalization Phase

Relocalization Phase — Template

To fill later:

- ▶ Online inputs
- ▶ Nearest-path logic
- ▶ Distance/heading outputs
- ▶ RViz visualization items

Validation

Validation — Template

To fill later:

- ▶ Real scenarios
- ▶ Simulation scenario
- ▶ Quantitative metrics
- ▶ Comparison criteria

Discussion

Discussion and Conclusion — Template

To fill later:

- ▶ Main findings
- ▶ Current limitations
- ▶ Scope boundaries
- ▶ Next steps

Conclusion

Thank you for listening!

Any questions?

Joao Pedro Martins do Lago Reis | Yann Kelvem da Silva Ramos
Record, Map, and Relocalize with AprilTags